KaxaNuk
Sharing knowledge

INVESTMENT BOOTCAMP

CREATE YOUR RESEARCHER AND LEARN TO USE THE INVESTMENT LAB.

Every investor holds convictions. Few can test them. KaxaNuk gives you the tools to validate them.

SESSION 01 · NOVEMBER 2026

KN

STOP BELIEVING.
START VALIDATING.



OUR FOUR SOLUTIONS GROW OUT OF ONE EIGHT-STEP PROCESS.

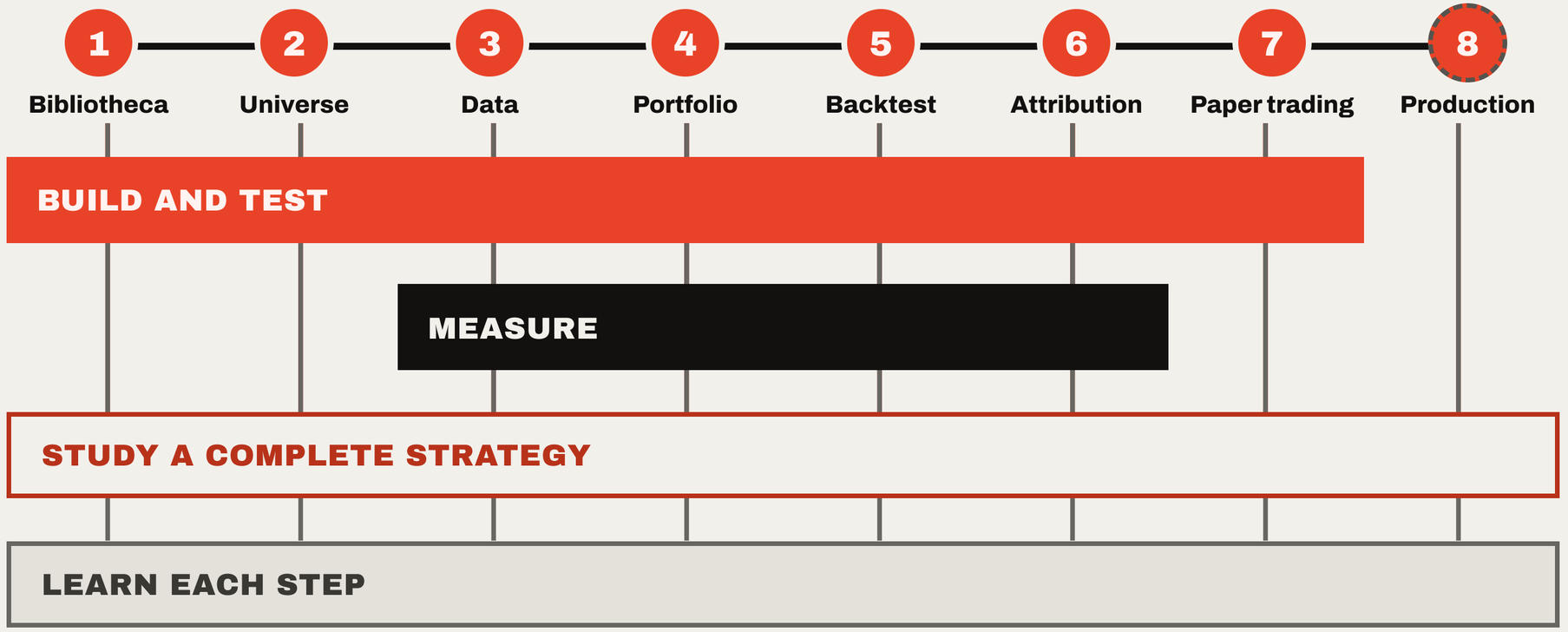
THE PROCESS
From a written belief to production.

INVESTMENT LAB
Build and test the idea, with clicks or with code.

ANALYTICS FACTORY
Measure every result against benchmarks, factors and risk.

RESEARCH MODELS
Complete strategies with code, to study or adapt.

TRAINING WORKSHOPS
Learn each step with real examples.



YOUR RESEARCHER BRINGS THE PROCESS TO EVERY USER.

UNIFIES

One conversation for the Lab, Analytics, Research Models and the workshops.

GUIDES

Walks you through every step of the process and every tool, whether or not you code.

CONNECTS

Your library, your strategies and your projects, in files you own.

THREE STEPS TO CREATE YOUR RESEARCHER.

1

OPEN AN ASSISTANT

In an empty folder, signed in.

2

PASTE ONE LINE

The README's, and send it.

3

ABOUT YOU

Who you are, what you do, its name.

github.com/KaxaNuk/Researcher

Open source · Install section

Scan the code, open the README and follow it with me: about ten minutes from start to finish.

OPEN AN ASSISTANT, SIGNED IN.

RECOMMENDED

CLAUDE CODE
Anthropic · Claude. The app's Code tab, no terminal.

CODEX
OpenAI · GPT. The Codex app, or the CLI.

COPILOT CLI
GitHub · Claude, GPT or Gemini. The CLI.

CURSOR
Cursor · several models. The editor.

ANTIGRAVITY
Google · Gemini. The CLI that replaced Gemini CLI.

~~A WEB CHAT~~
It can't run commands, so it can't install it.

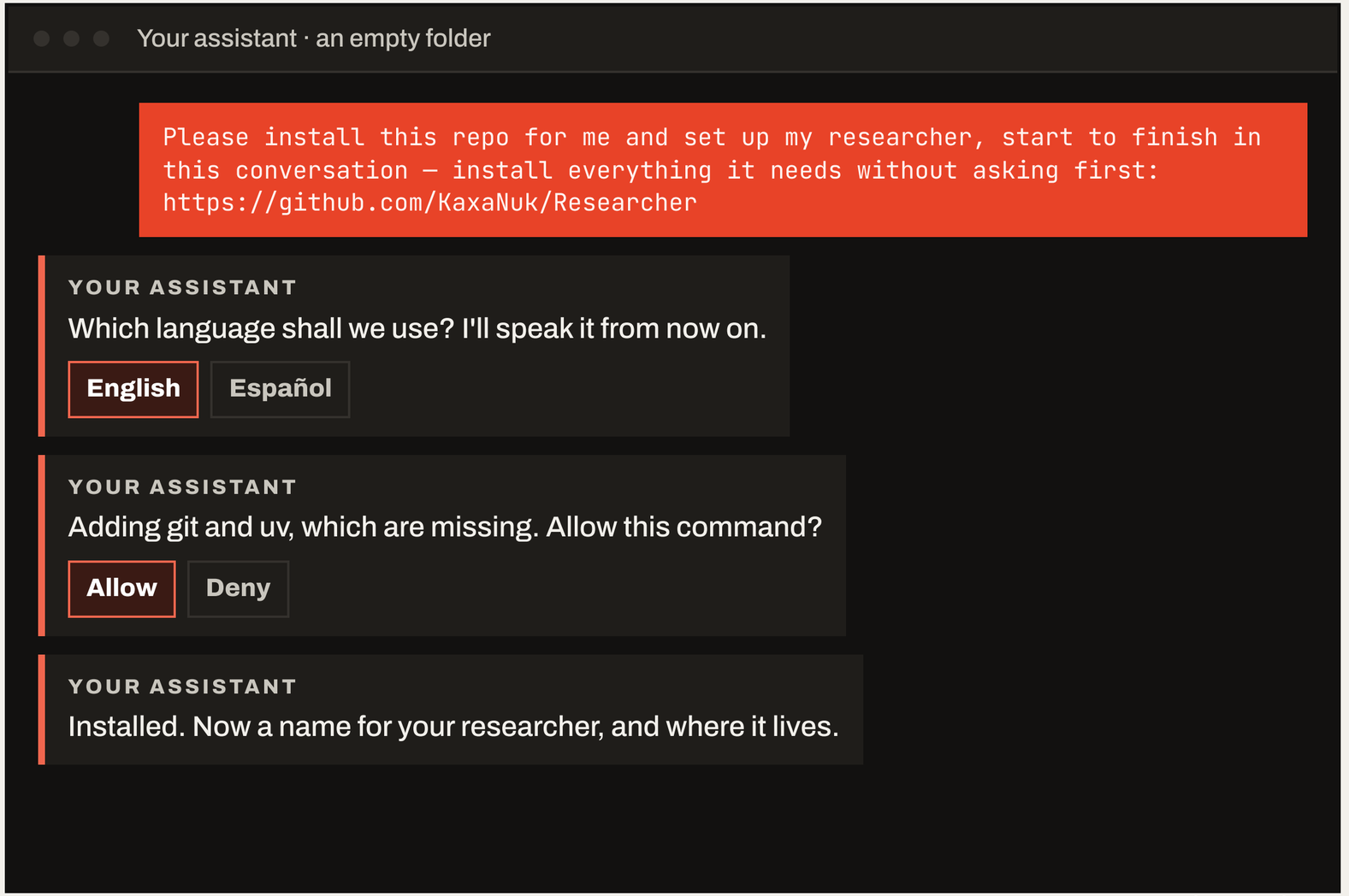
Tip: choose Local, not Cloud, an empty folder and a mode that asks before each command: Manual in Claude, Ask for approval in Codex.

COPY ONE LINE. SEND IT.

1. Copy the README line; paste it; send.
2. Read each command; allow it.
3. Answer its questions.

It asks your language, adds **git** and **uv** if missing and installs the researcher. If it stops: **SETUP.md**, in the repository.

git saves every version of your files · **uv** installs Python and its packages

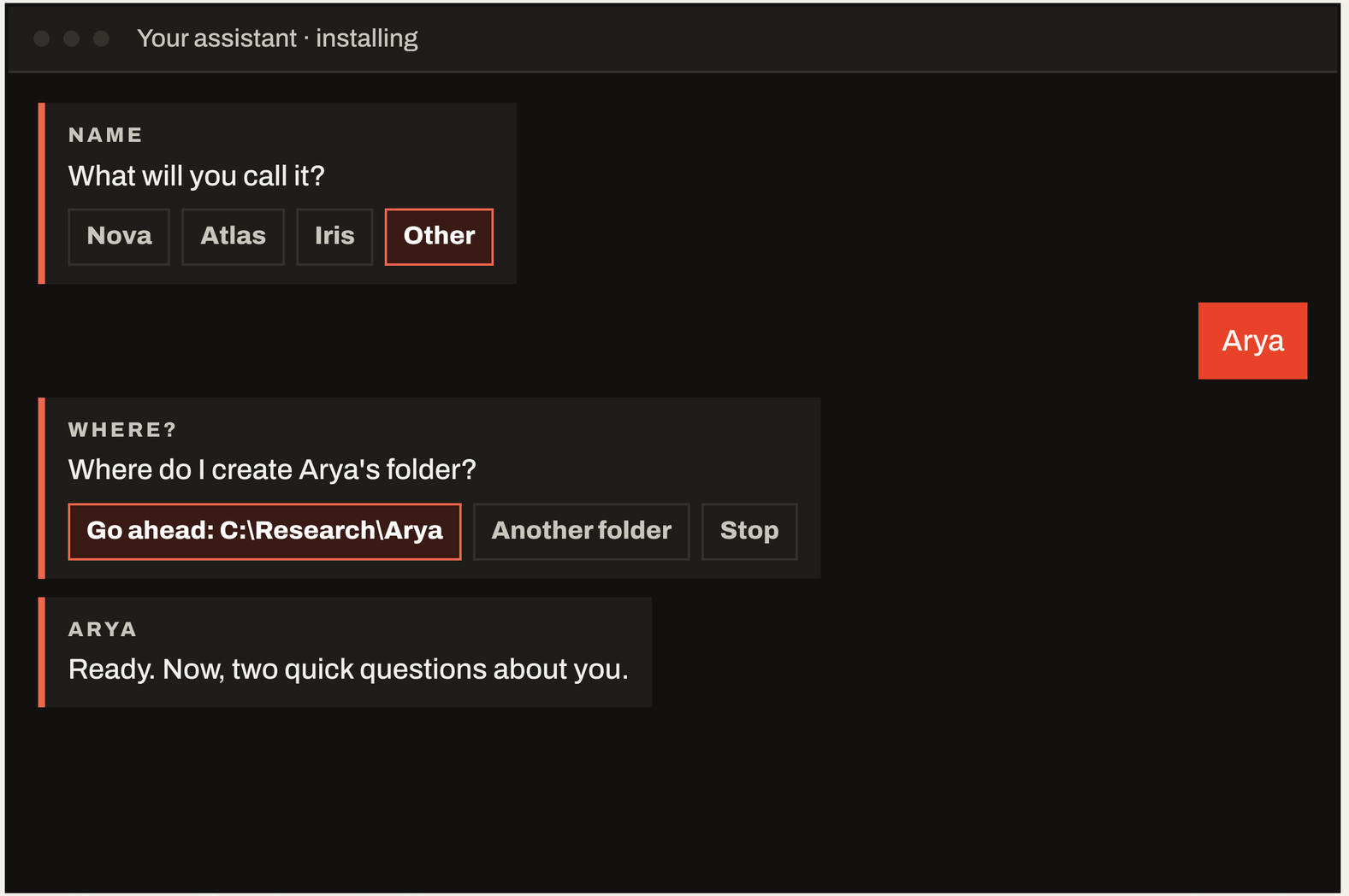


Your app still asks permission for every command

NAME IT. CHOOSE WHERE IT LIVES.

1. Name it: a suggestion, or your own.
2. A short path for its folder.
3. Then, two questions.

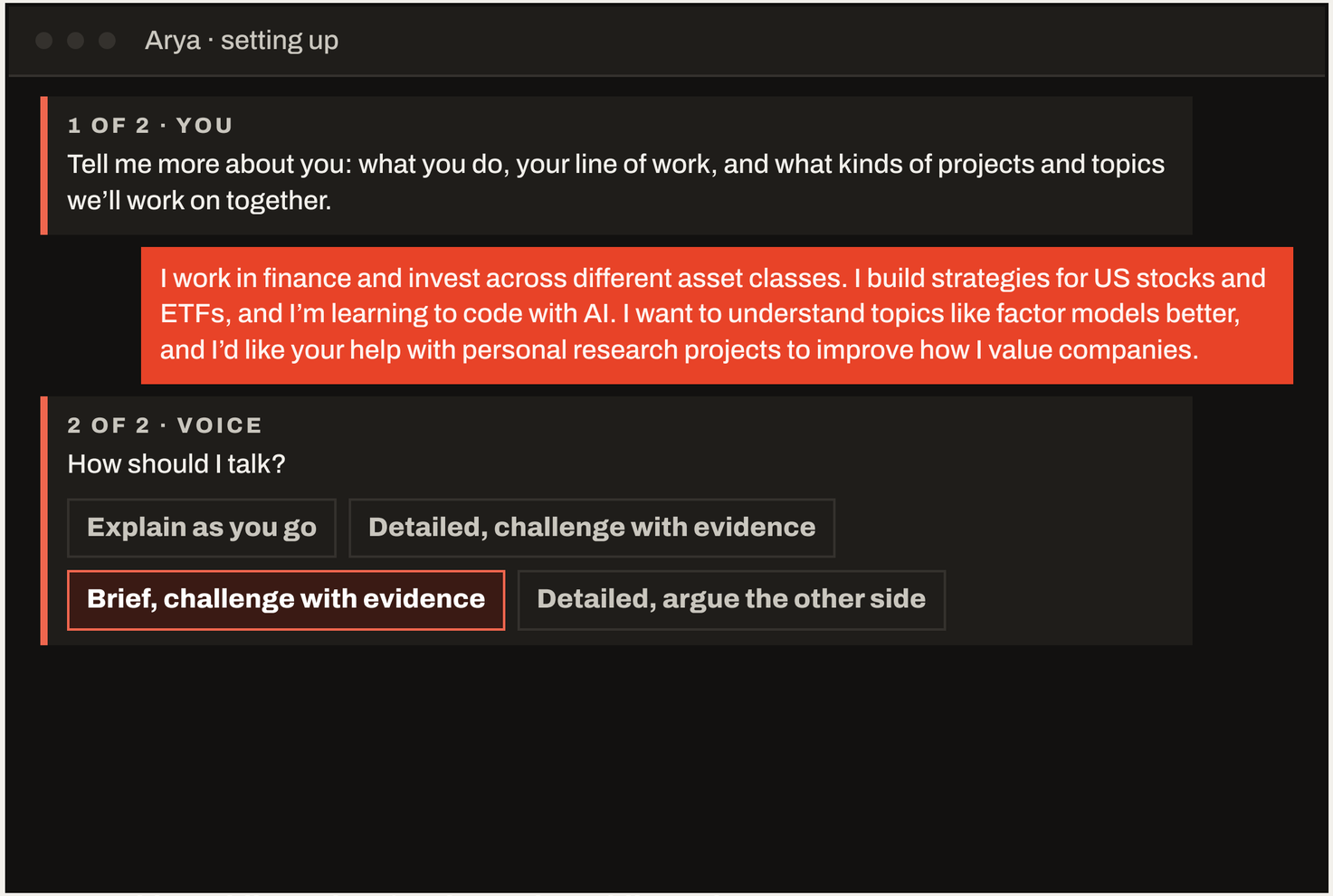
Call it by name in any folder. For research and learning: it can be wrong, and **every decision is yours**.



On Windows it suggests C:\Research\<name>, or D:\Research\<name> if you have a D: drive

YOU FIRST, THEN ITS VOICE.

About three minutes, and nothing about markets. “I’m just starting” is an answer too.



RESEARCHER.md

Works for: a finance professional who invests across asset classes

Here for: US stock and ETF strategies; learning to code with AI; personal research projects

Domains: US stocks, ETFs, factor models, company valuation

Voice: brief, challenge with evidence

It reads this first in every conversation. Fix anything wrong with a word, or edit it by hand.

A PLACE FOR EVERYTHING, AND EVERY FOLDER WITH A PURPOSE.

YOU WRITE IT · YOUR RESEARCHER WRITES IT · BOTH

RESEARCHER.md
READ FIRST

You, its voice and your rules.

Philosophy/
READ FIRST

How you work and how you invest, in your words.

Sources/

What you give it: papers, books, clippings.

Knowledge/

Its notes, each citing its source.

Studies/

Decisions weighed, for and against.

Lessons/

Lessons and their quizzes.

`Sources/` → read → `Knowledge/` → query · study · teach → `Studies/` · `Lessons/` → YOU DECIDE

IT'S ALL MARKDOWN

.md · formatted plain text

Any assistant and any editor can read it: no app lock-in.

An LLM sees the structure in the text, without guessing, and with few extra words.

git saves every change, and you can compare it line by line.

All in files you own, on your computer

TEACH IT WITH WHAT YOU HAVE READ.

1. **Save your documents in Sources/: books, papers, clippings.**
2. **Ask Arya to read it.**
3. **Approve its plan: it writes a cited note.**

Arya · its folder

ARYA
Save in Sources/ what you want me to know, or attach it here and I copy it.

Arya, read Sources/Papers/Markowitz_1952.pdf

ARYA
Which question is it for? Here is my plan. Do you approve?

go ahead

ARYA
Saved. From now on I answer from what you read, and cite the page.

Its library

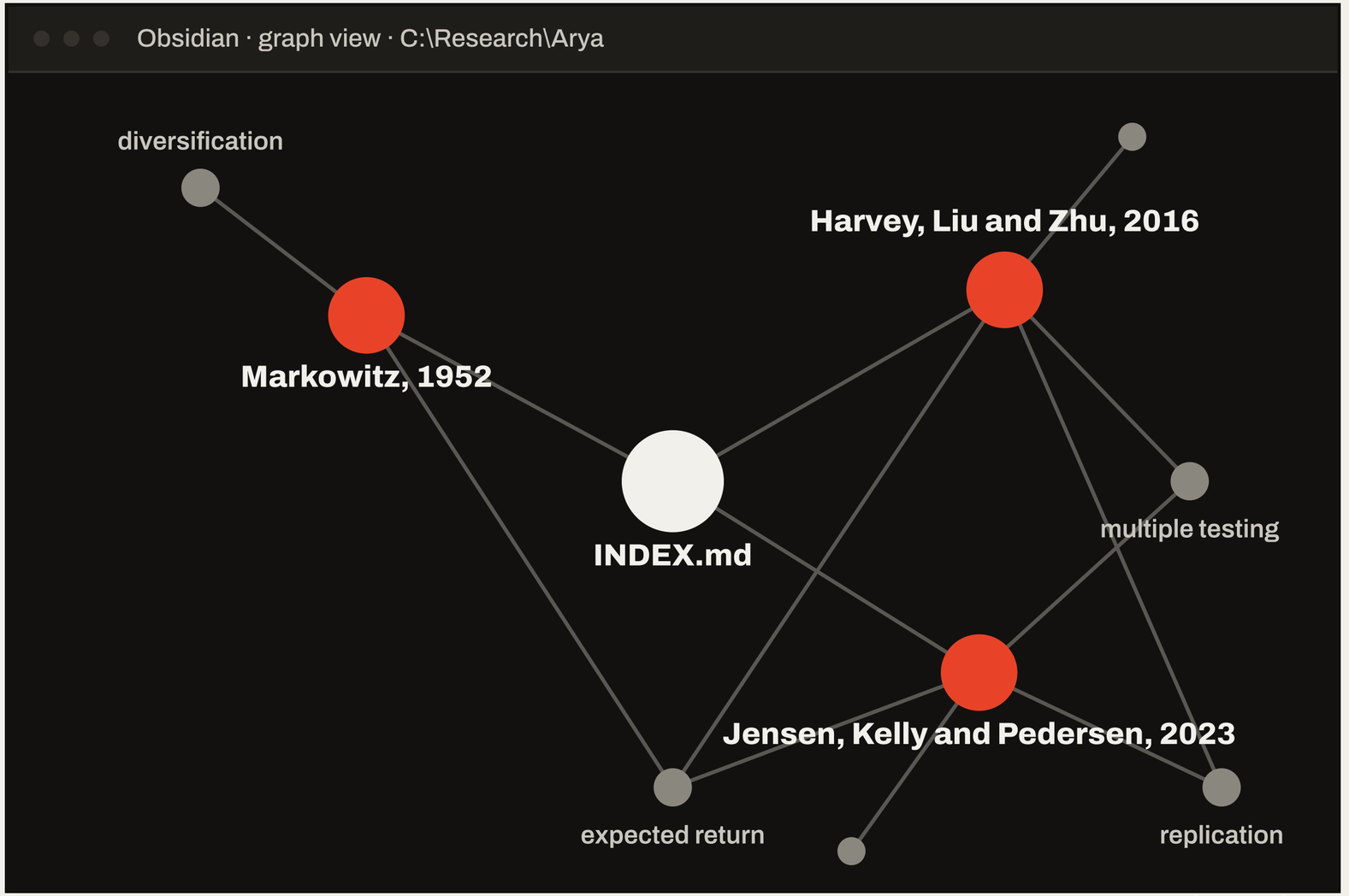
```
Sources/
   Books/
   Papers/
   Clippings/
   any folder you like/ •
   Papers/Markowitz_1952.pdf •
Knowledge/Finance/
   Markowitz_1952_….md •
   concept pages •
Knowledge/INDEX.md •
```

The folders are yours: add any you like, and read walks them all. What it reads, it cites.

WATCH YOUR KNOWLEDGE GROW.

1. Open its folder as a vault.
2. Follow links and backlinks; search.
3. Edit RESEARCHER.md by hand.

Each note links its **concept pages** and INDEX.md lists them all: the graph grows with every paper. Without code, it’s the easiest way to read and edit what it writes.



SEVENTY YEARS OF INVESTING IDEAS.

1

1952
MARKOWITZ
RISK WITH RETURN
Weigh risk with return.
Portfolio Selection

1964
SHARPE
A PRICE FOR RISK
Only market risk is paid: beta.
Capital Asset Prices

1968
JENSEN
THE FIRST TEST
Most funds did not beat the market.
The Performance of Mutual Funds

1970
FAMA
THE NULL
Prices reflect what is known.
Efficient Capital Markets

1979
KAHNEMAN & TVERSKY
THE PEOPLE
Losing hurts more than winning feels good.
Prospect Theory

1980
GROSSMAN & STIGLITZ
WHY AN EXCESS RETURN MUST EXIST
Costly information: someone is paid.
On the Impossibility of Informationally Efficient Markets

1993
FAMA & FRENCH
FACTORS
Size and value, beyond beta.
Common Risk Factors

1993
JEGADEESH & TITMAN
MOMENTUM
What rose tends to keep rising, for months.
Returns to Buying Winners and Selling Losers

1997
CARHART
THE FOURTH FACTOR
Fund streaks: factors and costs, not talent.
On Persistence in Mutual Fund Performance

1997
SHLEIFER & VISHNY
WHY IT PERSISTS
Seeing it is not holding it.
The Limits of Arbitrage

2004
LO
THE SYNTHESIS
Strategies crowd, fade, and come back.
The Adaptive Markets Hypothesis

2016
MCLEAN & PONTIFF
DECAY
Smaller once published.
Does Academic Research Destroy Stock Return Predictability?

2

2016
HARVEY, LIU & ZHU
TOO MANY TESTS
Most factors may be false: t above 3.0.
... and the Cross-Section of Expected Returns

3

2023
JENSEN, KELLY & PEDERSEN
THE REBUTTAL
Most do replicate.
Is There a Replication Crisis in Finance?

HOW IT EVOLVED
From risk and return to factors, and today's debate.

THEREFORE
① is your first read; then ② and ③ argue it out.

A SKILL FOR EVERYTHING YOU WANT TO LEARN OR TEST.

A skill is a manual your researcher opens when the task calls for it: the steps, the rules and the traps. It's how KaxaNuk's know-how travels, and stays current.

A command is the name you type to ask for one: `read`, `study`, `teach`. Without one, it also knows the skill from what you ask.

YOUR LIBRARY

`read` · `query` · `audit` · `refine` · `refresh-index`

Reads, answers with citations, finds gaps.

YOU AND WHAT YOU LEARN

`working-philosophy` · `investing-philosophy` · `teach` · `study` · `workshop`

How you work and invest, lessons and workshops.

YOUR STRATEGIES

`init-example` · `init-strategy` · `objective` · `blueprint` · `challenge` · `next`

Golden-Flow to study, and your own.

THE LAB

`universe` · `curator` · `analyzer` · `portfolio` · `backtest` · `attribution` · `alpha-decomposition` · `paper trading`

One skill per step, through paper trading.

DAY TO DAY

`brief` · `security` · `backup` · `deck` · `update` · `init-python-library`

Your brief with its weekly check, your keys kept safe and a copy on GitHub.

COMING SOON

`valuation` · `and what you ask for`

New skills to learn the process and the products, based on what our users ask for.

Ask it "what can you do?" to see them all · they arrive with every update

WHY NOT AN ASSISTANT ALONE? YOUR RESEARCHER RUNS IN IT AND ADDS WHAT IT DOES NOT KEEP.

	AN ASSISTANT ALONE	YOUR RESEARCHER
Your library	Forgets what it read.	What you read becomes notes on your computer; its answers cite them and say what's missing.
You	Starts over in every chat.	Your questions, your voice and your rules, in your own files.
Portability	Switch assistant or computer and you start over.	Claude Code, Codex, Copilot, Cursor or Antigravity, another account or computer: install it and it keeps everything, with a private copy on GitHub.
KaxaNuk's knowledge	Knows what anyone knows.	Skills for the eight-step process and each Lab library, with their traps.
Your own skills	Only brings its own.	Create the skills you need with the same architecture, in its folder, and they travel with it.

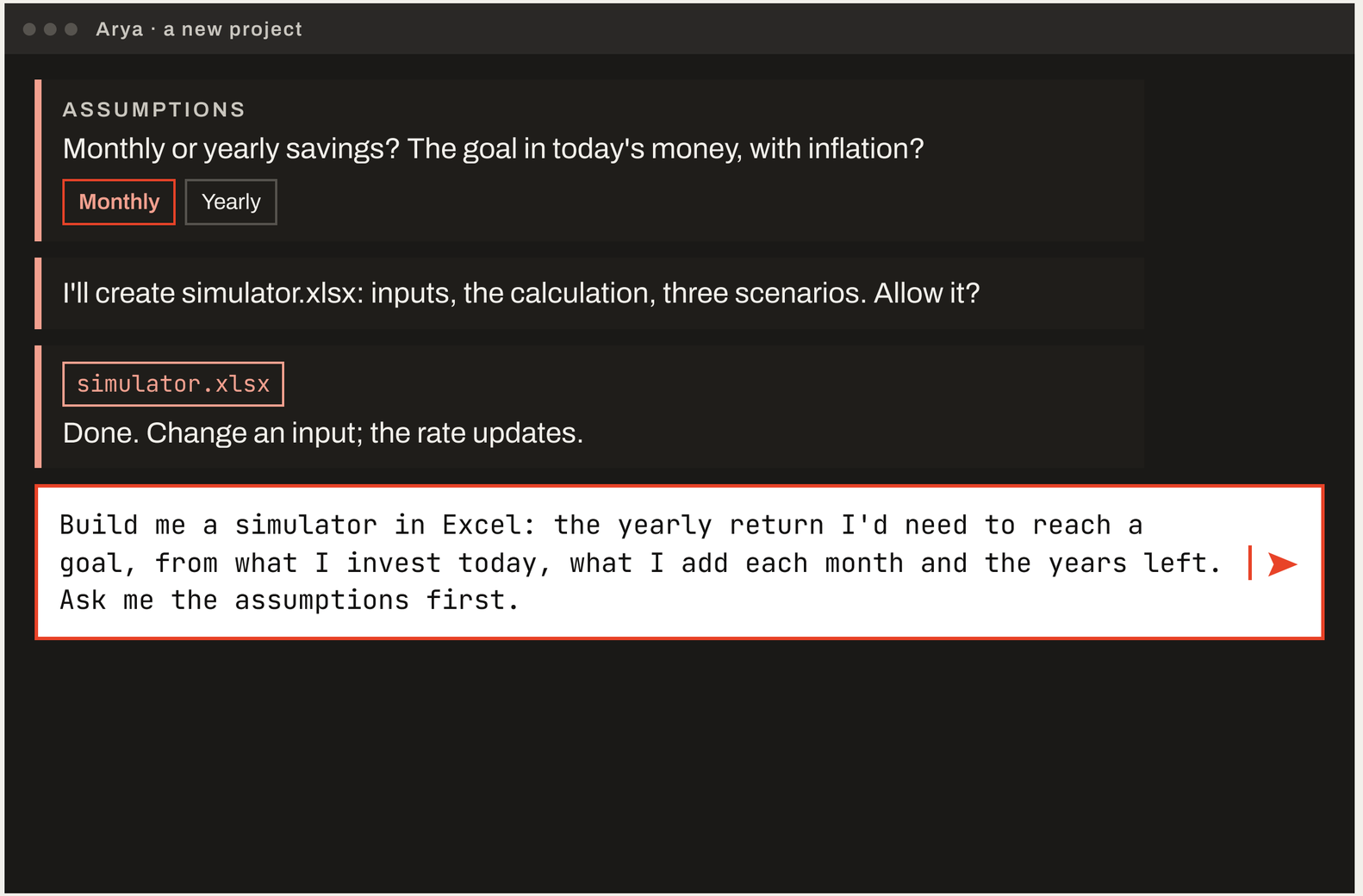
How much would your savings need to earn each year?

You invest $100,000 today and $5,000 every month, for 30 years. Your goal: $10 million when you retire.

Write your number before the next slide.

ASK FOR A TOOL. TRY IT WITH YOUR NUMBERS.

1. Ask for it; it asks the assumptions first.
2. Allow the file; open it.
3. Try your numbers: today, monthly, years, goal.



FOR $10 MILLION, 9.1% A YEAR.

Teaching computation, before inflation and fees.

9.1%

YOUR GOAL: $10 MILLION

$100,000 today, $5,000 a month, 30 years.

5.5%

A $5 MILLION GOAL

With the same saving.

11.7%

SAVING $2,500 A MONTH

For the same $10 million.

THE BOOTCAMP'S QUESTION Which strategy would earn it, without fooling yourself?

HOW YOU GET TO 9.1%, AND WHAT RAISES IT.

Teaching computation: it says how much you would need, never how much you will earn.

STEP	WHAT IS COMPUTED	RESULT
1 · What you put in	$100,000 today and 360 contributions of $5,000.	$1,900,000
2 · What the return adds	$10 million less what you put in.	$8,100,000
3 · The rate that gets there	**The one that takes what you put in to $10 million in 30 years.**	**9.1% a year**
4 · With fees	1% a year on your balance: the rate before they are charged.	10.2%
5 · With inflation	4.9% a year, Mexico's from 2015 to 2025; the goal in today's money and your contributions rising with it.	14.5%
6 · With both	Fees and inflation together.	15.7%

Balance = 100,000·(1+r)^30 + 5,000·((1+i)^360 − 1)/i, with i = (1+r)^(1/12) − 1 · inflation: INEGI's INPC, from 89.047 in Dec 2015 to 143.042 in Dec 2025, read 2026-11-03 · the fee, an assumption.

COPY THE PROMPT. USE IT FOR ANY GOAL.

1. **Copy the prompt; paste it in your assistant.**
2. **Open the Excel; change your numbers.**
3. **Run the dashboard and fit it to your goal.**

Your assistant · a new folder

`simulator.xlsx`

Done: the Excel, and the Dash dashboard.

Run it with uv run app.py and change it as you like: it is your tool.

```
I want a savings simulator for a goal: my retirement or any project I need to save for.
Inputs, all editable: what I have today, what I add each month or each year, the years left,
the goal, and the inflation and fees I assume. Outputs: the yearly return I would need to get
there; or, if I fix the return, how much I would have to add or how many years it would take;
a year-by-year table and a chart of the balance against the goal. Give me two versions: 1)
simulator.xlsx, inputs in coloured cells and the formulas in view; 2) a Python project with
uv: the calculation in a module with tests, and a Dash dashboard, easy to fill in, that runs
with one command. Before writing, ask me the assumptions you are missing and show me your
plan. Every number is a teaching computation: how much I would need, never how much I will
earn. Do not recommend what to invest in.
```

Works in Claude Code, Codex, Copilot, Cursor or Antigravity · teaching computation: how much you would need, never how much you will earn.

YOUR RESEARCHER UNITES THE LAB PIPELINE: ONE SKILL PER STEP.

1 **Bibliotheca**
`read · objective · blueprint`

2 **Universe**
`universe`

3 **Data**
`curator · analyzer`

4 **Portfolio**
`portfolio`

5 **Backtest**
`backtest`

6 **Attribution**
`attribution · alpha-decomposition`

7 **Paper trading**
`the graduation gate`

8 **Production**
`outside the Lab`

You

Which step am I on?

Your researcher

Step 5, the backtest. Plan: one Backtest Engine run with its skill. Approve?

`init-example` · the full Golden-Flow, to study

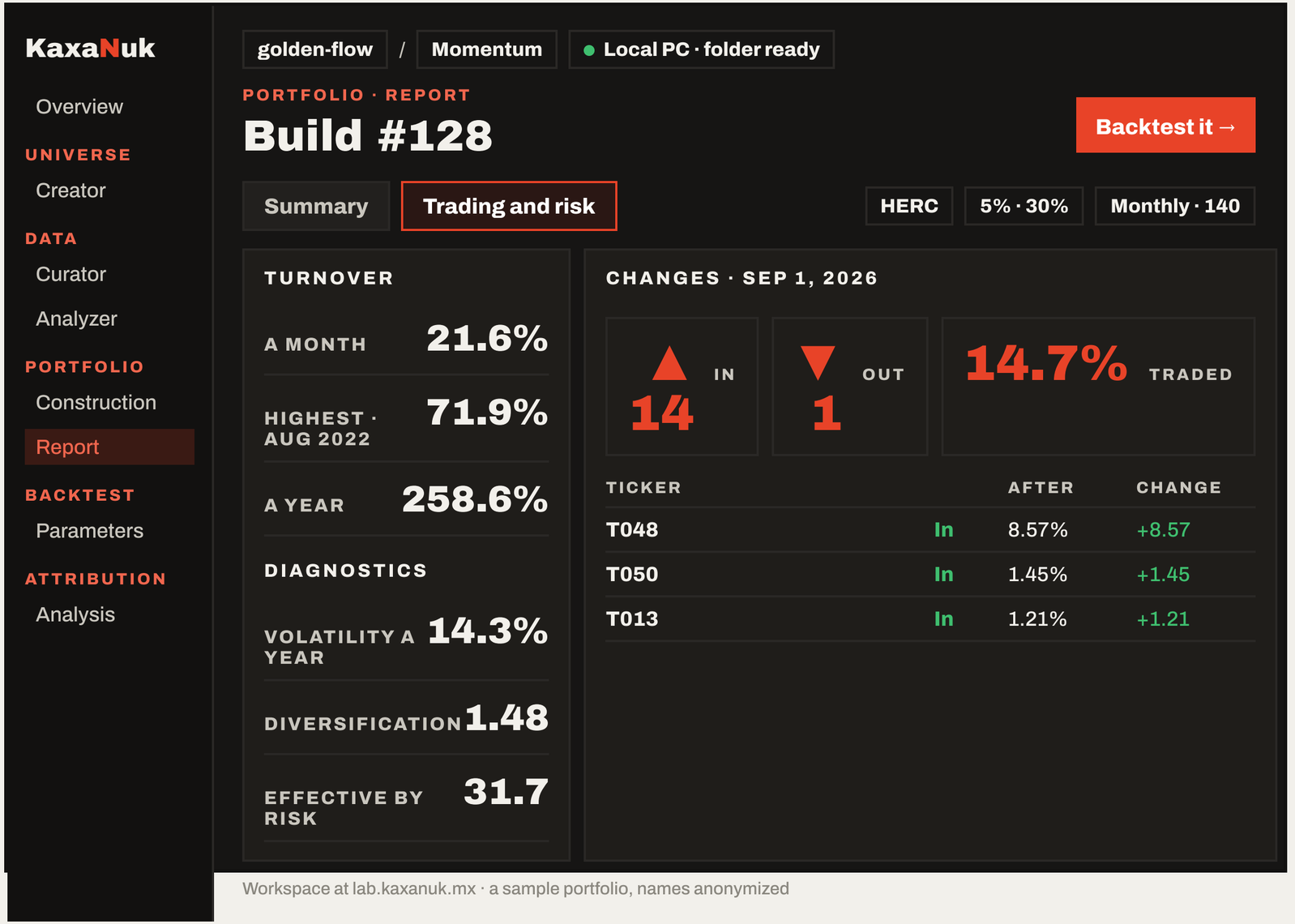
`init-strategy` · your own, with the same folders

Knows your step, what's next and which library to use · plan and approval before writing · the numbers come only from the Lab

ONE LAB, WITH CLICKS OR WITH CODE.

A library is tested code that does one step of the process. The workspace and your code use the same six, in Python.

- **Data Curator**
- **Data Refinery**
- **Data Analyzer**
- **Portfolio Construction**
- **Backtest Engine**
- **Attribution Analysis**



Workspace at lab.kaxanuk.mx · a sample portfolio, names anonymized

IN CODE, EVERY NUMBER IN SIGHT.

TRANSPARENCY

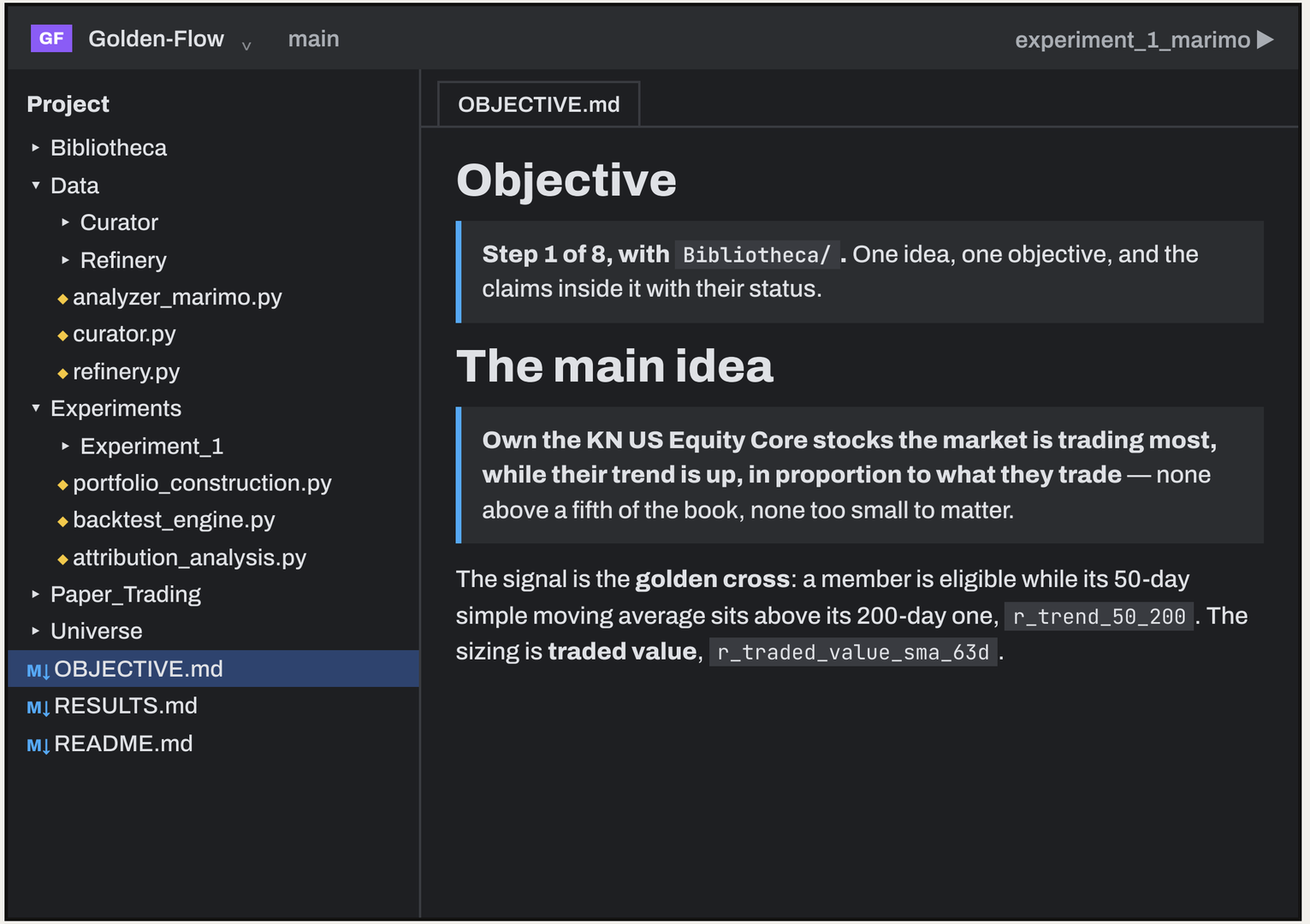
Each step is a file you read; each number, a calculation you can follow.

DEBUG AND CUSTOMIZE

Line by line, in PyCharm, VS Code or whatever editor you use; build any idea the platform doesn't cover.

ANY PROVIDER

The Data Curator connects Sharadar, Yahoo Finance, FMP, LSEG or your own data: CSV, Parquet, pandas.



Golden-Flow in PyCharm · the same engine as the platform

WITH CLICKS OR WITH CODE: THE SAME SIX LIBRARIES, ONE FLOW.

THE SIX LIBRARIES
The same in the workspace and in your code.

THE PLATFORM
The workspace, with clicks.

ANALYTICS FACTORY
Attribution against the KN benchmark.

YOUR RESEARCHER
It guides you through the eight steps.

THE PROCESS
Eight steps: it knows where you are.

YOUR CODE
Each step, a file.

RESEARCH MODELS
Golden-Flow, and yours.

OUR BENCHMARK AND FACTOR MODEL EXPLAIN WHERE RETURNS COME FROM.

702–915 members a day in the KN US Equity Core **2,198** securities priced, 2000–2026

GOLDEN-FLOW · RETURN BY FACTOR, 2000–2026



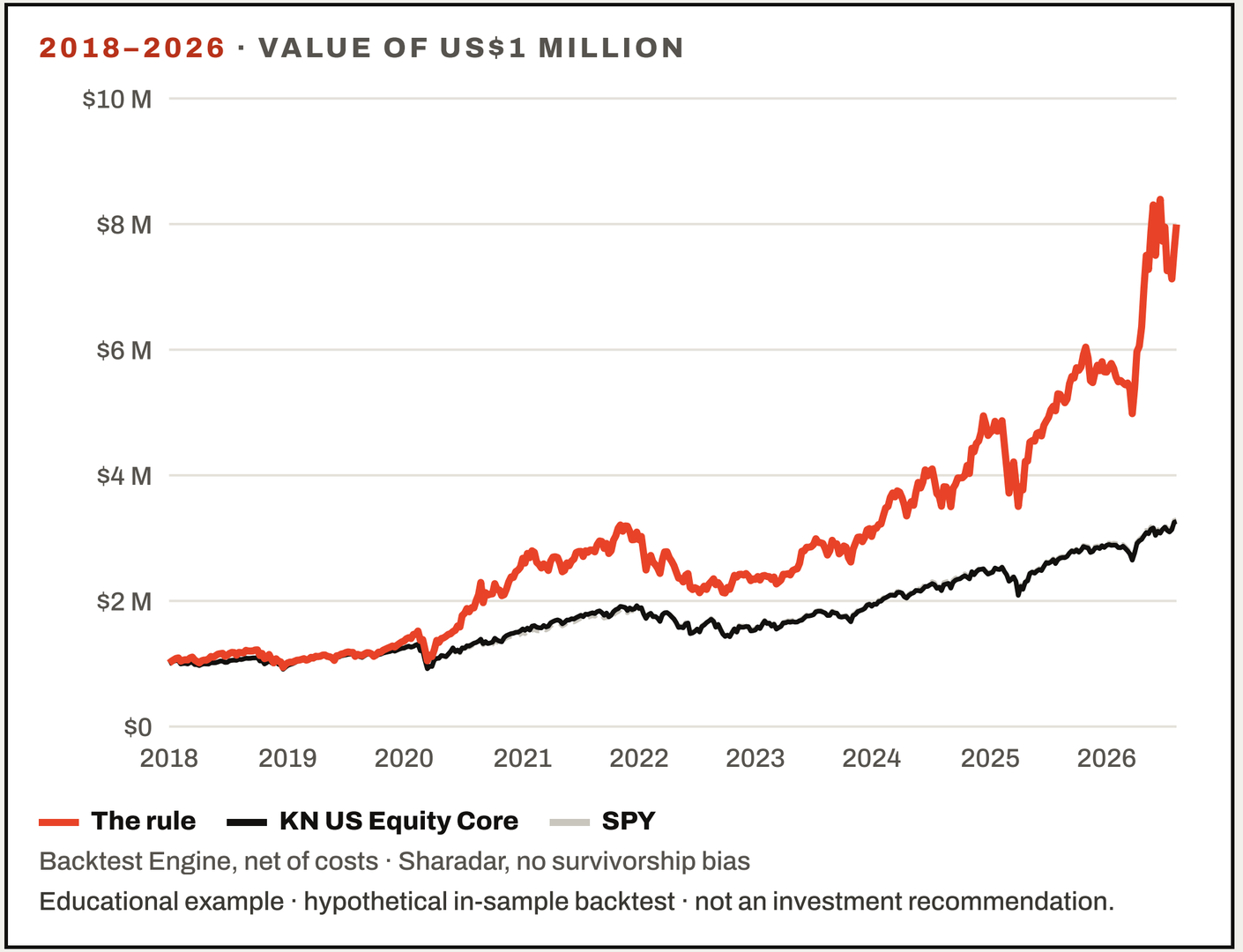
Return points by factor, daily KN factor model. In red, the idiosyncratic: what no factor explains. Attribution Analysis, hypothetical backtest.

GOLDEN-FLOW: IN ITS PERIOD, THE RULE WORKS.

THE RULE

The most traded stocks in the KN US Equity Core, while their trend is up, weighted by traded value.

Sharpe KN Core: **0.579**	**0.819**
Annual return KN Core: **14.19%**	**26.24%**



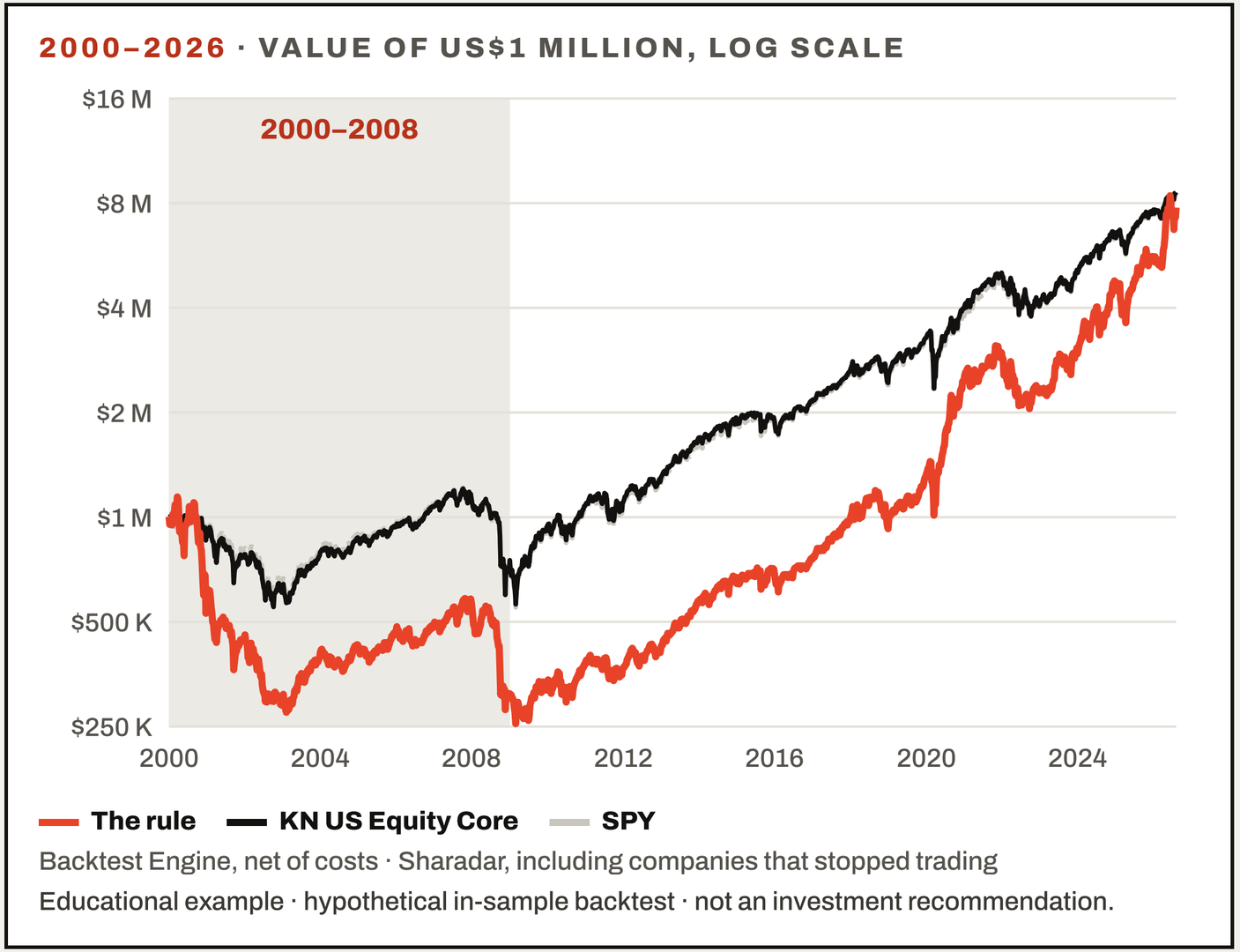
Backtest Engine, net of costs · Sharadar, no survivorship bias

Educational example · hypothetical in-sample backtest · not an investment recommendation.

THE SAME RULE, SINCE 2000.

Sharpe KN Core: **0.315** · SPY: **0.313**	**0.221**
Max drawdown KN Core: **–53.9%**	**–78.4%**
2000–2008, annual KN Core: **–3.41%**	**–11.71%**

On 2015–2026 data, an earlier version passed the gate; on 26 years, it doesn't. A short period doesn't prove a signal always works: the test is how many regimes it survives. It's sound and regime-dependent: it goes to paper trading on probation, 12 months, with its review registered before day one.



Backtest Engine, net of costs · Sharadar, including companies that stopped trading

Educational example · hypothetical in-sample backtest · not an investment recommendation.

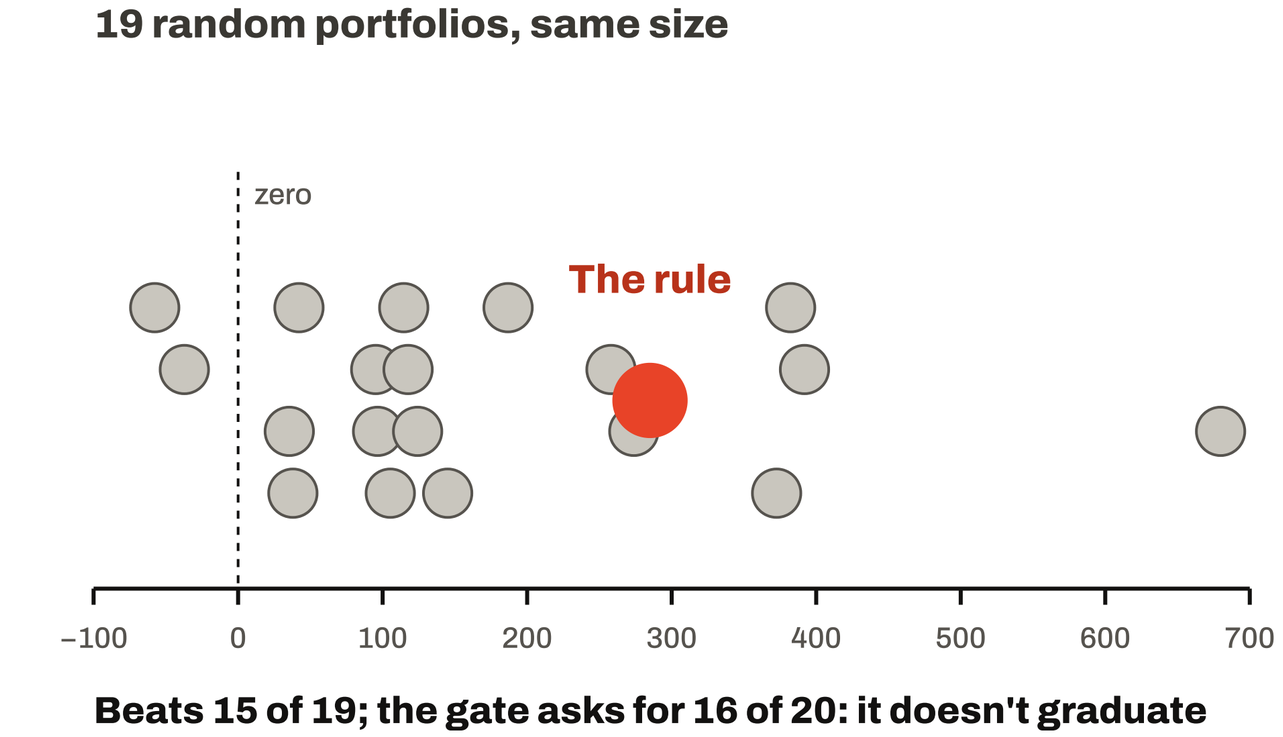
WHERE THE RETURN CAME FROM, AND WHAT SANK IT.

Beta. It cost 207.6 points over 26 years: the rule held the most market-sensitive stocks.

The idiosyncratic. +285.0 points, what no factor explains, and 85.0 from technology.

Against chance. The rule's idiosyncratic return beats 15 of 19 random portfolios; the gate asks for 16 of 20.

Against random portfolios



Idiosyncratic return, 2000–2026 · random portfolios of the same size, fully priced

Educational example · hypothetical in-sample backtest · not an investment recommendation.

WHAT ATTRIBUTION TELLS US TO TEST NEXT.

BETA IN DOWNTURNS

Over 26 years beta cost 207.6 points. Test a rule that cuts beta exposure when the market falls.

ITS OWN EXPERIMENT

RESIDUAL VOLATILITY

Cost 47.8 points over 26 years. Test capping names with extreme residual volatility.

ITS OWN EXPERIMENT

SECTOR CONCENTRATION

Technology added 85.0 points and concentrates the risk. Test a cap per KN sector.

ITS OWN EXPERIMENT

BEFORE

The hypothesis and what would refute it, written before the rule.

CONTROL

The same rule with a single ingredient removed.

TRIALS

152 portfolios run and counted, published with the result.

NET

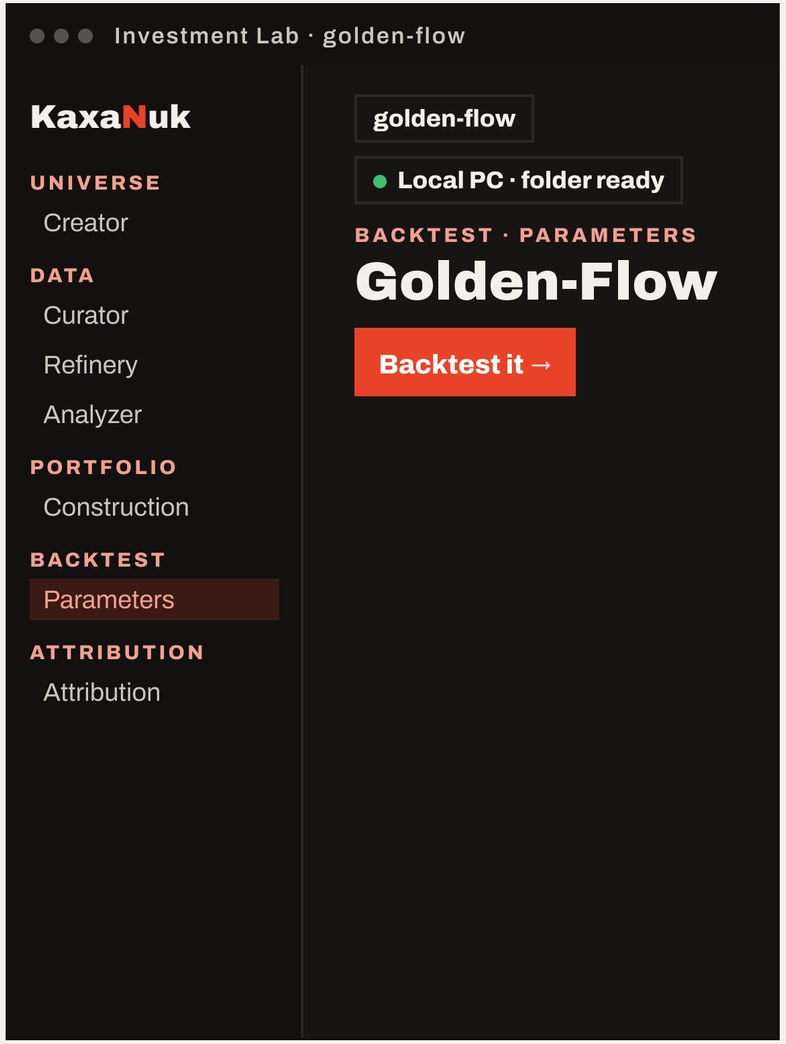
Commissions, slippage and interest on cash, inside the engine.

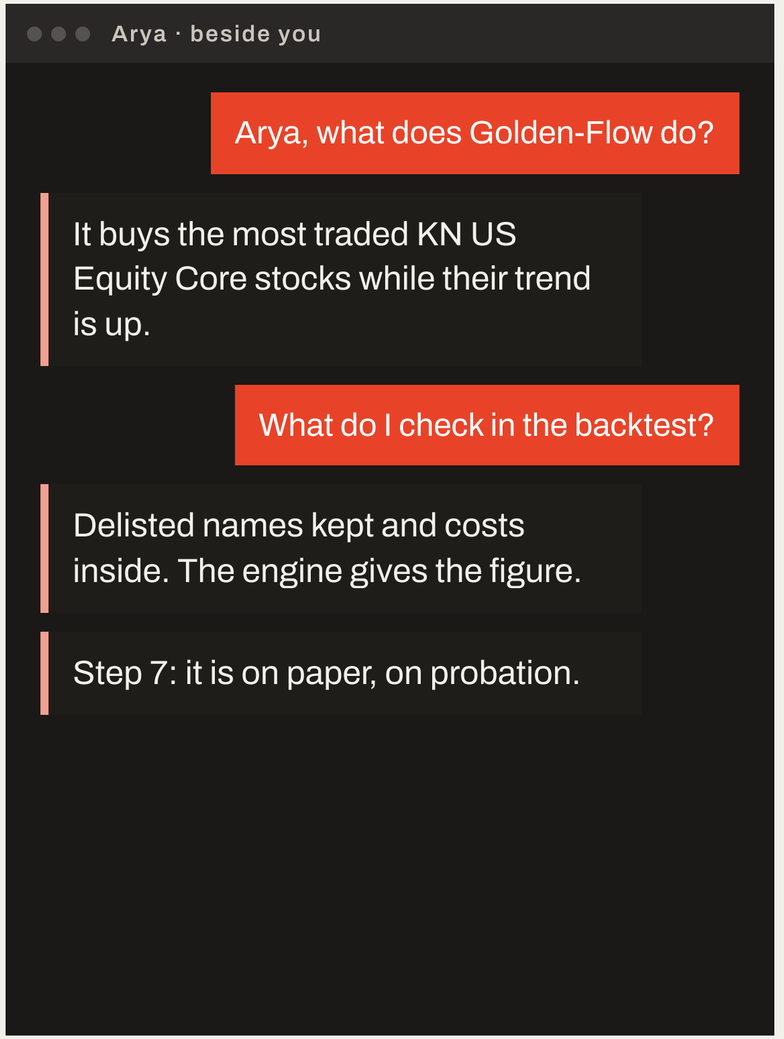
NOW, THE LAB LIVE.

One strategy from idea to attribution: with clicks in the workspace and in code, with the researcher alongside.

A WHOLE STRATEGY, WITH CLICKS.

1. Ask it what the strategy does.
2. Walk each step with clicks; Arya says what to check.
3. From the objective to paper trading, no code.

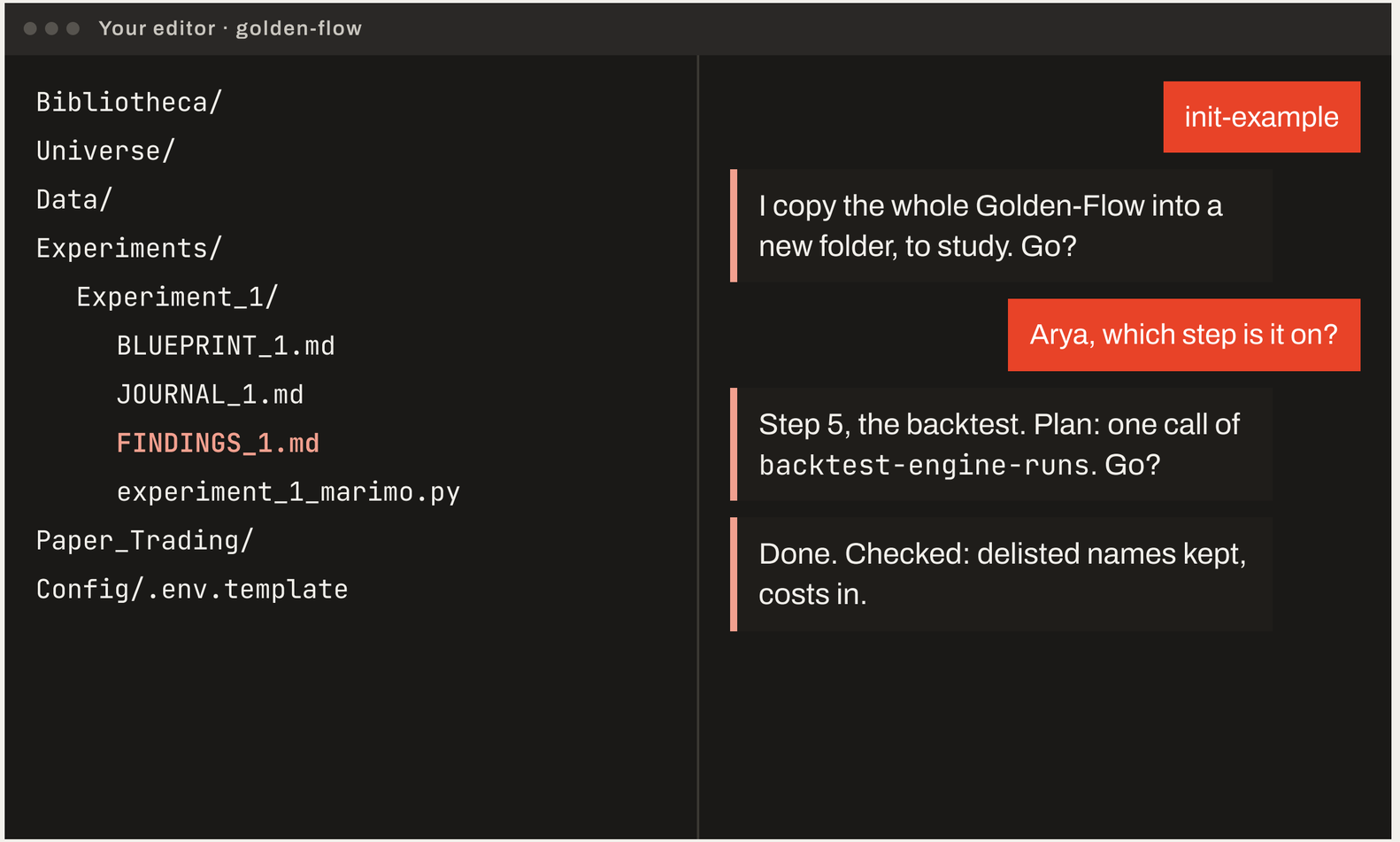




Workspace at lab.kaxanuk.mx, by subscription · Golden-Flow is an example to study, never a recommendation.

THE SAME, IN CODE: INIT-EXAMPLE, AND ARYA GUIDES YOU.

1. init-example copies the whole strategy.
2. It knows the step and uses each library.
3. Guards the common mistakes.



IDEAS ARE SIMPLE. TESTING THEM WELL IS NOT.

IDEA	THE RULE	WHAT IT TESTS	ITS CONTROL	BENCHMARK
Bonds or cash	Hold long bonds while the yield curve steepens; else cash.	Does the curve warn ahead of bonds?	Always hold the bonds.	Cash.
USD/MXN	Hold dollars while USD/MXN is above its 200-day average; else pesos.	Does the peso's trend persist?	Always hold pesos.	Pesos.
Stocks: Golden-Flow	The most traded KN US Equity Core stocks, while their trend is up.	Does the trend add to liquidity?	The same, trend ignored.	KN US Equity Core.
Stocks: value with growth	The cheapest by price to earnings, only if their earnings grow; rebalanced each quarter.	Does growth tell cheap from broken?	The cheapest, without the growth filter.	The whole market.
Portfolio: 60/40	60% stocks and 40% bonds, rebalanced each year.	Does diversifying cut risk without costing return?	The same, never rebalanced.	Stocks alone.

THE LAB MAKES IT SIMPLE a backtest without the common mistakes, and an attribution that separates what your rule adds from factor exposure. Ideas and hypotheses, validated with evidence, not beliefs.

START YOURS: INIT-STRATEGY.

1. **Tell it your idea; it suggests names.**
2. **Pick the name; approve its plan.**
3. **Open it in a new session; follow its SETUP.md.**

Arya · a new strategy

init-strategy

Tell me the idea in a sentence or two: what it would buy, and why you think it works.

Stocks with a rising trend and growing earnings.

Here are some names. The name is yours.

trend-earnings | momentum-earnings | Other

Ready in C:\Research\trend-earnings, its first version saved. Open it and follow its SETUP.md.

Your strategy

```
trend-earnings/
  OBJECTIVE.md .
  Bibliotheca/ .
  Universe/ .
  Data/ .
  Experiments/ .
  Paper_Trading/ .
  SETUP.md .
```

The same template as Golden-Flow · we write its OBJECTIVE.md in S02.

BACK UP YOUR RESEARCHER AND YOUR STRATEGIES.

Your researcher · its home

- **A private copy on GitHub** backup
- read: Mangram EXAMPLE
- The interview EXAMPLE
- First version EXAMPLE

Your strategy · its repository

- **A private copy on GitHub**
- Experiment 1: findings EXAMPLE
- Experiment 1: blueprint EXAMPLE
- First version EXAMPLE

Why: a new computer, a mistake undone, your whole history.

THE SESSION'S IDEAS, IN FOUR ANSWERS.

01

WHAT MAKES YOUR RESEARCHER YOURS?

Your files: your voice, your rules and your library, on your computer.

02

WHAT DOES YOUR SIMULATOR SAY, AND NOT SAY?

How much you would need to earn for your goal, with fees and inflation. Not what earns it.

03

HOW DO YOU TRAIN IT?

Documents in Sources/ and `read`: every note cites its source.

04

WHAT DOES THE LAB DO, STEP BY STEP?

Eight steps, one skill at each, with clicks or with code.

Next, S02: from idea to signal, with your objective, universe and data.

PLAY WITH ALL ITS SKILLS.

01
YOUR PHILOSOPHY

How you work, then how you invest.

`working-philosophy · investing-philosophy`

02
A LESSON

Anything, with a quiz.

`teach`

03
YOUR BRIEF

Daily, checked weekly.

`brief setup`

04
A STUDY

An idea.

`study`

05
YOUR PORTFOLIO

Holdings and rules.

`Portfolio/`

06
AND MORE

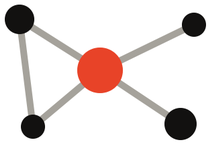
A deck; your own library.

`deck · init-python-library`

MISSING A SKILL? Open an issue at github.com/KaxaNuk/Researcher.

More arrive with each version: `update` brings them.

BEFORE SESSION 02: ARRIVE READY.



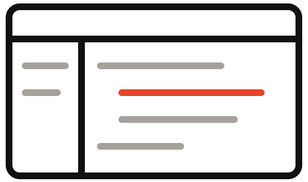
01 TEACH IT MORE

- Give it more to read: books, papers, clippings.
- Try three of its skills.
- Ask it what you need; `teach` teaches you.



02 YOUR SIMULATOR

- Try your own numbers, with and without inflation.
- Keep your goal: it is S02's objective.



03 YOUR FIRST STRATEGY

- `init-strategy`: create it, and name it.
- Your idea in one line, for your goal.
- `init-example`: browse Golden-Flow beside it.
- First Steps: CF P03, MF P01, FM P06.

YOU HAVE YOUR RESEARCHER. NOW TEST YOUR IDEAS.

Train it every day, take it to every project, and validate your ideas with evidence.

LEARN

Then Coding Foundations, Financial Markets and Mathematical Finance.

`github.com/KaxaNuk/Investment-Bootcamp`

TEST

Your strategy on the Lab; universes and factors from Analytics.

`kaxanuk.mx/lab` · `kaxanuk.mx/analytics`

TELL US

What your researcher helped you do.

QUESTIONS `workshops@kaxanuk.mx`

NEXT

Session 02 · From Idea to Signal

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